

VISTRA - Technical Documentation

Implementation reference for perception, risk mapping, localization, planning, safety, dashboard and verification

Evidence policy: PS-specific VISTRA content is from the supplied blueprint (B1). External technical and dataset facts are tied to the numbered source register. Project performance numbers are not claimed until measured.

1. Implementation contract

The implementation should preserve module boundaries so the same navigation code can run in simulation, recorded video and optional physical hardware.

Interface	Inputs	Outputs
CameraInterface	Frames + timestamps	RGB/stereo frame + quality
VehicleInterface	Linear/angular command	Wheel/vehicle state
SensorInterface	IMU / wheel / auxiliary	Measurement + timestamp
Perception	Image	Class posteriors + dynamic tracks
Localization	Images + inertial/odom	Pose + covariance + health
Planner	Risk map + pose + goal	Path + (v,w)
Safety	Trajectory + risk + obstacles	Limit / stop state
Dashboard	Telemetry/events	Human-readable state

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

2. SegFormer-B0 deployment path

SegFormer is used for terrain semantic segmentation. The original paper presents a hierarchical transformer encoder with a lightweight MLP decoder. Hugging Face documentation lists the MiT-B0 variant at 3.7M parameters.

Deployment flow: dataset adapter -> preprocessing -> SegFormer inference -> per-class probabilities -> uncertainty -> class mapping -> BEV projection. ONNX Runtime is the proposed portable inference runtime.

Stage	Engineering control
Preprocessing	Fix input size, normalization and crop policy before benchmarking
Inference	Export once; version the ONNX file
Postprocessing	Retain probabilities for uncertainty; do not keep only argmax
Benchmark	Measure FPS and full-pipeline latency on actual target hardware
Integrity	Never copy paper FPS into the VISTRA result table

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | **S4** SegFormer paper | **S5** Hugging Face SegFormer documentation | **S12** ONNX Runtime documentation

3. Risk-map implementation

For each pixel, compute class-probability traversability and normalized entropy. Project the evidence into a ground-referenced BEV cell. Combine terrain cost, uncertainty, geometry and proximity.

Step	Operation
1	Get posterior p_k for each pixel
2	Compute $T = \sum p_k t_k$
3	Compute normalized entropy U
4	Project to BEV
5	Fuse temporal evidence with decay
6	Apply geometry and proximity terms
7	Inflate cost using pose covariance
8	Publish map to planner and dashboard

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

4. Stereo VO / SLAM interface

Use a provider interface so ORB-SLAM3 or RTAB-Map can be swapped without changing the rest of the stack. The fallback can use OpenCV feature matching, PnP and RANSAC.

Signal	Stored field	Why
Features	tracked_features	Visual health
Inliers	ransac_inliers	Geometric health
Reprojection	reproj_error	Pose quality
Covariance	pose_cov_trace	Uncertainty
Quality	blur/brightness/coverage	Input quality
Status	tracking_state	Normal / degraded / lost

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | S6 ORB-SLAM3 paper

5. Global planning: D* Lite

Represent the BEV map as a cost grid. Mark red/lethal cells as impassable. D* Lite can reuse search information as edge costs change. That fits the VISTRA risk map, which changes as new frames arrive.

The planner should publish a path plus a reason when no path is available. Repeated failure escalates to rotate-search, backtracking or safe stop.

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | **S7** D* Lite - Carnegie Mellon Robotics Institute

6. Local planning: DWA

DWA evaluates reachable linear/angular velocity pairs. The VISTRA scoring function should expose goal progress, risk, smoothness and clearance. Store weights in configuration and log them with each benchmark run.

Reject any velocity whose simulated trajectory intersects a lethal risk cell or violates configured vehicle/motor limits.

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | **S8** Dynamic Window Approach - Carnegie Mellon Robotics Institute

7. TTC implementation

The TTC module forward-simulates the vehicle footprint using the current (v,w). For tracked moving objects, use position and velocity under the simple constant-velocity model defined by B1.

Parameter	Unit	Source / status
t_latency	s	Measured project value; blank until benchmark
v	m/s	Current planner command
a_max	m/s^2	Vehicle/simulator parameter
tau_stop	s	Computed = t_latency + v/a_max
TTC	s	First predicted footprint overlap

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

8. NCS and recovery state machine

NCS should be calculated at a documented control cadence. Hysteresis should use separate entry and exit thresholds to avoid rapid state switching.

State	NCS range	Behavior
NORMAL	> 0.7	Nominal speed cap
CAUTION	0.4 - 0.7	Reduce speed; increase inflation
RECOVER	0.2 - 0.4	Stop/creep; relocalize
SAFE_STOP	< 0.2	Hold + alert

Every transition is an event object containing timestamp, previous state, new state, trigger, signals and action. The dashboard explanation feed and benchmark log should read from the same event.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

9. ROS 2 + Gazebo integration

Use ROS 2 topics/services/actions only at module boundaries. Gazebo Harmonic provides the simulated world, vehicle dynamics and sensor streams. Current official Gazebo documentation lists Harmonic as an LTS release with May 2029 EOL.

A deterministic scenario generator should fix the seed and generate obstacle positions, lighting, start and goal. Save the exact scenario configuration for replay.

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | **S2** ROS 2 Jazzy documentation | **S3** Gazebo Harmonic documentation

10. Dashboard data contract

Field	Type	Example purpose
timestamp	float	Ordered telemetry
frame_id	int	Traceability
ncs	float	Behavior gate
perception_confidence	float	Subsystem health
localization_confidence	float	Subsystem health
pose	x,y,yaw	Map
pose_covariance	matrix/summary	Risk inflation
risk_map	grid	Planning
path	polyline	Route
ttc	float	Safety
state	enum	NORMAL/CAUTION/RECOVER/SAFE_STOP
cmd_v/cmd_w	float pair	Planner command
wheel_left/right	float pair	Actuator command
decision_log	event	Explanation

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

11. Software verification

Test	Acceptance criterion
Probability mapping	Known posterior vectors yield expected T and risk direction
Entropy	One-hot gives near zero; uniform posterior gives maximum normalized entropy
Projection	Calibration points land in expected ground coordinates
Wheel mapper	v-only gives equal wheels; turn follows configured sign convention
TTC	Synthetic overlap produces finite TTC; clear path gives no event
NCS hysteresis	No rapid state toggling around threshold
Recovery	Each injected fault produces documented transition + event log
Planner safety	No commanded trajectory traverses lethal cells

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

12. Release and audit rules

- Version models, configuration and scenario seeds.
- Keep raw run logs, not only aggregate charts.
- Record software commit hash in benchmark output.
- Keep real evaluation data isolated from training.
- Measure FPS and end-to-end latency on the actual target machine.
- Do not present the dashboard starter synthetic values as system results.
- Document all limitations and known failure modes.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

Appendix - Trusted Source Register

B1 - Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

PS 26126 metadata, VISTRA design, formulas, roadmap, benchmark protocol, risks and limitations.

Supplied project source PDF.

S1 - Bharat Electronics Limited - Official Homepage

Official BEL organizational description.

<https://bel-india.in/homepage/>

S2 - ROS 2 Jazzy documentation

ROS 2 Jazzy documentation and packages.

<https://docs.ros.org/en/jazzy/>

S3 - Gazebo Harmonic documentation

Official Gazebo Harmonic documentation and release information.

<https://gazebosim.org/docs/harmonic/>

S4 - SegFormer paper

Original SegFormer architecture and efficiency rationale.

<https://arxiv.org/abs/2105.15203>

S5 - Hugging Face SegFormer documentation

Model configuration; MiT-B0 listed at 3.7M parameters.

https://huggingface.co/docs/transformers/main/model_doc/segformer

S6 - ORB-SLAM3 paper

Visual and visual-inertial SLAM, stereo support and multi-map behavior.

<https://arxiv.org/abs/2007.11898>

S7 - D* Lite - Carnegie Mellon Robotics Institute

Incremental heuristic replanning for unknown terrain.

<https://publications.ri.cmu.edu/d-lite>

S8 - Dynamic Window Approach - Carnegie Mellon Robotics Institute

Velocity-space local collision avoidance using robot dynamics.

<https://publications.ri.cmu.edu/the-dynamic-window-approach-to-collision-avoidance>

S9 - RUGD paper / official host

7,453 pixel-wise labeled images; unstructured outdoor driving challenges.

https://rugd.vision/pdfs/RUGD_IROS2019.pdf

S10 - RELIS-3D official repository

13,556 LiDAR scans, 6,235 images, 20-class ontology and multimodal data.

<https://github.com/unmannedlab/RELIS-3D>

S11 - RELIS-3D paper

Off-road semantic scene understanding dataset motivation and benchmarks.

<https://arxiv.org/abs/2011.12954>

S12 - ONNX Runtime documentation

Cross-platform ML inference runtime.

<https://onnxruntime.ai/docs/>

S13 - CVAT documentation

Visual dataset annotation and management platform.

https://docs.cvat.ai/docs/getting_started/overview/

S14 - OpenCV documentation

Computer-vision implementation APIs.

<https://docs.opencv.org/>

S15 - FastAPI documentation

Python API framework used for dashboard integration.

<https://fastapi.tiangolo.com/>